Assay Document Normalization — Post-Fix Audit

Date: 2026-04-16 **Scope:** Re-audit of `assay_definitions` after the BIMS write-path fix. This report focuses narrowly on bringing the 32 outlier documents into structural agreement with the 206-document majority.

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1. Current state

Signature	Doc count	Status
#1 — canonical legacy	**206**	Treat as the target shape
#2 — clinical assays	3	Outlier (different `BCODE` shape + extra clinical fields)
#3 — `useSingleCost` added	5	Minor outlier (one extra field)
#4 — `useSingleCost` + empty `lockedBy`	24	Minor outlier (two extra fields, one is a stray empty object)
Total	**238**	**32 docs need normalization**

Re-running the BIMS-shape scanner (`scripts/find-bims-assays.ts`) confirms zero documents bear the BIMS-creation fingerprint — every one of the 238 still carries the legacy markers (`A########` ID, `BCODE` , `hidden` , `protected`). The write-path changes are dormant until someone actually creates an assay through BIMS. Nothing in mongo has been altered.

What "match the majority" means

The 206-doc canonical shape (signature #1) is exactly:

```
keys: BCODE, __v, _id, corrections, createdAt, description, duration,
      hidden, isActive, name, protected, reagents, updatedAt, versionHistory
```

with `BCODE` shaped as `{ deviceParams: {...}, code: [...] }` . The 32 outliers diverge from this on a handful of specific keys. The fix per cluster is below.

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2. Per-signature normalization plan

Signature #2 — clinical assays (3 docs)

Affected docs:

- `A1A2B3C4` — "CRP Quantitative"
- `A5E6F7D8` — "IL-6 Quantitative"
- `A9B0C1D2` — "TNF-alpha Semi-Quantitative"

Divergences from canonical:

1. `BCODE` is a **bare array** — should be wrapped as `{ deviceParams: DEFAULT, code: <array> }` .
2. Has 7 extra fields not in the canonical shape: `cartridgeLife` , `diluent` , `functions` , `lotLife` , `matrix` , `version` , `useSingleCost` .

Recommendations (decision required):

Field	Sample value	Recommendation
`BCODE` (array)	`[{command:"START",…},…]`	**Wrap** in `{ deviceParams: DEFAULT_DEVICE_PARAMS, code: <array> }`. Required for shape parity.
`cartridgeLife`	`180`	**Decision needed** — drop, or migrate to `shelfLifeDays` (already in schema)?
`lotLife`	`365`	**Decision needed** — drop, or stash under a new `lifecycle: { lotLife, cartridgeLife }` block?
`diluent`	`"PBS-T"`	**Decision needed** — discard, or move to `metadata.diluent`?
`matrix`	`"Serum"`	Same as `diluent`.
`functions`	`{ standard: {...}, cutoff: {...} }`	**Likely worth preserving** — looks like a calibration/scoring spec.
`version`	`"1.0"`	Drop. The schema's `versionHistory.length` already serves this purpose.
`useSingleCost`	`false`	Drop (it's the schema default — see signature #3 discussion).

The clinical commands inside the `BCODE` array (`START` , `DELAY` , `MOVE` , `READ` , `FINISH`) **also differ** from the canonical Title-Case vocabulary (`Start Test` , `Delay` , `Move Microns` , `Read Sensor` , `Finish Test`). Bringing these to full firmware-compatible parity is a separate effort — wrapping the array gets the *structure* right but not necessarily the *content*. Flag for firmware review before promoting these 3 to production use.

Signature #3 — extra `useSingleCost` (5 docs)

Affected docs:

- `AA812E34` — "Exp 256 6 -> 7"
- `A3DAD9AE` — "B code transition the big yank COPY"
- `A0A85E21` — "Gen 5 Optical Scan Test"
- `A11A0CFE` — "Bait Assay"
- `AB024599` — "Bait Assay COPY"

Divergence from canonical: has the field `useSingleCost: false` .

Recommendation: Leave as-is. `useSingleCost` is declared in the current Mongoose schema with `default: false` , so it's a legitimate field that just happens to have been written explicitly on these 5 docs. Removing it would only hide it from the doc; the value would still be `false` to any reader. If absolute key-set parity is desired, run `$unset: { useSingleCost: 1 }` against these 5; otherwise no action.

Signature #4 — extra `useSingleCost` + empty `lockedBy` (24 docs)

Divergences from canonical:

1. `useSingleCost: false` — same as signature #3. Same recommendation: optional `$unset` .
2. **`lockedBy: {}`** — an empty object on every one of these 24. Misleading: it implies the assay was locked by someone, but `lockedAt` is absent on all 24, so they were never actually locked. The empty `{}` is a

Mongoose schema artifact — when the schema declares `lockedBy: { _id: String, username: String }`, Mongoose can instantiate it as an empty subdoc on `.create()` even when no value is supplied.

Recommendation: `$unset` the empty `lockedBy` on all 24 docs (and drop `useSingleCost` if you want full parity). This is safe — none of these were ever locked, and a reader checking `if (assay.lockedAt)` would already give the right answer; the empty `lockedBy` only causes problems for code that checks `if (assay.lockedBy)` (which would currently be true even though the assay isn't locked).

To prevent future occurrences, the `AssayDefinition` schema's `lockedBy` field should be wrapped or made conditional so Mongoose doesn't auto-instantiate it on `.create()`. The simplest fix is to declare it with a sub-schema and `_id: false`, or to use `Schema.Types.Mixed`.

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3. Suggested migration script

A single migration would handle all 32 outliers. Sketch:

```
// scripts/normalize-assay-shapes.ts (proposed — NOT YET WRITTEN)
const DEFAULT_DEVICE_PARAMS = {
  delayBetweenSensorReadings: 100, integrationTime: 128, gain: 0, ledPower: 300
};

// Sig #2: wrap bare-array BCODE
db.assay_definitions.find({ BCODE: { $type: 'array' } }).forEach(d => {
  db.assay_definitions.updateOne(
    { _id: d._id },
    { $set: { BCODE: { deviceParams: DEFAULT_DEVICE_PARAMS, code: d.BCODE } } }
  );
});

// Sig #4: drop empty lockedBy
db.assay_definitions.updateMany(
  { lockedBy: {}, lockedAt: { $exists: false } },
  { $unset: { lockedBy: 1 } }
);

// (optional) Sig #3 + #4: drop useSingleCost when it equals the default
db.assay_definitions.updateMany(
  { useSingleCost: false },
  { $unset: { useSingleCost: 1 } }
);

// Sig #2 clinical fields: handle individually based on the decisions in §2.
```

Same dry-run-first / `--apply` pattern we used for the equipment label migration. The script itself is not written yet — pending the field-by-field decisions in §2.

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4. Summary of recommended actions

Action	Affects	Risk	Decision-required?

Wrap signature #2's array `BCODE` as `{deviceParams, code}`	3 docs	Low	No
Decide fate of clinical fields (`cartridgeLife`, `lotLife`, `diluent`, `matrix`, `functions`, `version`)	3 docs	Low–Med	**Yes**
Re-translate signature #2's `BCODE` commands to canonical Title-Case vocabulary	3 docs	Med (firmware)	**Yes** — needs firmware review
`$unset` empty `lockedBy` on signature #4	24 docs	Very low	No
`$unset useSingleCost: false` (cosmetic key-parity)	29 docs	Very low	Optional
Tighten `lockedBy` schema declaration to prevent future stray `{}`	All future writes	Low	No

Once the decisions in §2 are made, the migration script can be written and run in one pass. Estimated end-state: a single signature for all 238 docs.

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5. What this report does *not* cover

This is a narrow normalization pass. The broader recommendations from `docs/assay-shape-report.md` (read-path fixes, dead-field cleanup, reagent backfill, firmware vocabulary lock-down, hidden/protected semantics, etc.) are still open and orthogonal to this normalization.